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ORIGINAL PAPER

Morphological and genetic characterization of Mount Kenya brush-furred rats (*Lophuromys* Peters 1874); relevance to taxonomy and ecology

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Abstract

Brush-furred rats (*Lophuromys* Peters 1874) are the most abundant and widely distributed small mammals on Mount Kenya. However, their ecology is poorly known and their taxonomic status is questionable. This study investigates whether or not the morphology and genetics of the Mt. Kenya *Lophuromys* correspond with elevation, vegetation, and climate variables. We also investigate the taxonomic relationship of the Mt. Kenya *Lophuromys* to the Aberdare Range and Mt. Kilimanjaro populations using genetic and craniodental data. Specimens were captured along Mt. Kenya's western and eastern slopes, and other specimens (from Aberdare and Mt. Kilimanjaro) were obtained from Museum collections. The datasets were analyzed using multivariate and univariate statistics, discriminant function analysis, and phylogenetic analyses. The results show that Mt. Kenya *Lophuromys* specimens from different habitats and elevations are significantly differentiated by external morphology but not craniodental variables, and have low genetic variation and no population genetic structure, supporting the entire population is consistent with a single species model. In addition, Mt. Kenya and Aberdare *Lophuromys* cluster separately from Mt. Kilimanjaro *Lophuromys* based on craniodental analyses while phylogenetic and haplotype analyses also support the Mt. Kenya and Aberdare *Lophuromys* as evolutionarily isolated from Mt. Kilimanjaro's population. These results contradict recent studies that hypothesized the distribution of *Lophuromys margarettae* in lower elevations of Mt. Kenya with *Lophuromys zena* restricted to the higher elevations.

Keywords Genetic diversity · *Lophuromys zena* · Morphological variation · Phylogeny · Taxonomy · Ecology

Introduction

Species of the genus *Lophuromys* Peters 1874 are a major component of the small mammal fauna in tropical sub-Saharan Africa forests (Dieterlen 1976; Musser and Carleton 2005). On Mount Kenya, *Lophuromys* is the most abundant and widely distributed small mammal genus (Musila et al. 2019), constituting a major component of the diet of small

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carnivores and raptors (Mwebi et al. 2019). The Mt. Kenya *Lophuromys* belong to the dark brush-furred rats (*Lophuromys aquilus*) species group (Verheyen et al. 2002) that are adapted to the high-rainfall forests of Eastern Africa where they are locally abundant and inhabit diverse niches (Dieterlen 1976; Dieterlen 2013). Recent taxonomic accounts resolve the Mt. Kenya *Lophuromys* population to two species, *L. zena* and *L. margarettae*, with the later distributed in lower elevations and the former in higher elevation habitats (Verheyen et al. 2002; Musser and Carleton 2005; Verheyen et al. 2007). The taxonomic history of the Mt. Kenya *Lophuromys* has been consistently disputed in past and contemporary accounts (Musser and Carleton 2005), and little is known of their ecologies despite several studies (Lavrenchenko et al. 2007; Verheyen et al. 1996, 2000, 2002, 2007).

A decade after Dollman (1909) described *Lophuromys zena* as new to science, Hollister (1919) discredited its species status due to anatomical resemblances to *L. aquilus*. However, Osgood (1936) affirmed that the anatomical features dismissed by Hollister were substantially distinct from *L. aquilus* to consider *L. zena* a valid species. Allen (1939) listed *L. zena* as a subspecies of *L. aquilus*, views which Misonne (1971) disagreed with and consequently considered *zena* as a synonym of *L. flavopunctatus*. Verheyen et al. (2002) resolved *L. zena* as a valid species occurring in higher elevations of Mt. Kenya and the Aberdare Range with lower elevations occupied by *L. margarettae*. Musser and Carleton (2005) endorsed the taxonomic validity and distribution of both *L. zena* and *L. margarettae* as resolved by Verheyen et al. (2002). Verheyen et al. (2007) affirmed the findings of Verheyen et al. (2002), further validating the species status and distribution of *L. zena* and *L. margarettae* on Mt. Kenya, and in the process elaborated that samples from Aberdare and Kaptagat represented *L. zena* while those from Mt. Elgon, Solai, and Cherangani were *L. margarettae*. They also argued that Mt. Kilimanjaro *Lophuromys* samples were a contiguous population of *L. aquilus*, quite unlike the case of Mt. Kenya. Carleton and Musser (2013), underscoring the taxonomic controversy of the genus *Lophuromys*, challenged the validity of both *L. zena* and *L. margarettae* as recognized in Musser and Carleton (2005) and proposed them as synonyms of *L. flavopunctatus*. In contrast, the subsequent taxonomic treatments of African rodents by Monadjem et al. (2015) and Denys et al. (2017) recognized *L. zena*, *L. margarettae*, and *L. aquilus* as valid species. Overall, these taxonomic conflicts are a hindrance to effective conservation as they fail to accurately determine the building blocks of biodiversity studies (i.e., valid "species") and consequently, the ability of conservation projects to correctly prioritize biodiversity hotspots (Mace 2004). Since the taxonomic disagreements regarding Mt. Kenya *Lophuromys* relate to morphological (external and craniodental) characterization mostly to the exclusion of genetic data, an approach that integrates morphology, genetics, and distributional data would offer not only integrative taxonomic assessment but also offer insights into their elevational ecology which might correlate with other montane small mammals.

In spite of several zoological surveys that have documented the diversity, abundance, and taxonomy of rodents, insectivores, and chiropterans across many Eastern Africa mountain ecosystems (Peterhans et al. 1998, 2008, 2010, 2013; Clausnitzer and Kityo 2001; Demos et al. 2014a, 2014b; Stanley et al. 2014; Demos et al. 2015; Stanley and Kihaule 2016; Sabuni et al. 2018), Mount Kenya has lacked new surveys focusing on small mammals until recently (Musila et al. 2019). Because of Mt. Kenya's high elevation (up to 5200 m), equatorial location, pronounced environmental elevational stratification, and insularity, resident species are presented with diverse opportunities for morphological and genetic diversification (Garcia-Porta et al. 2017) and habitat colonization (Hedberg 1969). The mountain's heterogeneous climate and topography across short elevational distances, typical of tropical mountains (Korner et al. 2017), and position in a matrix of arid lowland savannah and grassland ecologically isolate its fauna. In addition to these ecological factors, Mt. Kenya is faced with progressive local human habitat encroachment (Eckert et al. 2017) and globally changing climatic regimes (Beniston 2003) which are likely to further ecologically isolate its fauna and may facilitate intraspecific morphological and genetic differentiation. Small mammals, such as *Lophuromys*, which are the most abundant mammal taxa in most terrestrial ecosystems, can act as indicator species for habitat health assessing the ecological impact of climate change and human habitat disturbance (Landres et al. 1988; Moritz et al. 2008; Blois et al. 2010; Avenant 2011; Boutin and Lane 2014; Menéndez et al. 2014; Hope et al. 2017). Thus, taxonomic and ecological studies of small mammals are germane to the conservation of biodiversity. Understanding patterns of morphological and genetic variability of small mammals in unique ecosystems such as Mt. Kenya is therefore vital to ecosystem management to curb biodiversity loss (Hoffmann and Sgro 2011). Such applications are hindered within the Mt. Kenya ecosystem due to taxonomic uncertainty and limited ecological understanding of the resident small mammal fauna.

Morphological and genetic characterizations are undoubtedly the most common technique for species delimitation among mammalian taxa (Jones and Safi 2011). In mountainous ecosystems where complex geology and habitat fragmentation over short distances can facilitate elevated rates of adaptive and non-adaptive diversification leading to high levels of biodiversity and endemism (Hedberg 1969), interspecific morphological convergence is common, especially among small mammals (Michaux et al. 2005; Garcia-Porta et al. 2017). In effect, morphology might be uninformative in small mammal taxonomic and ecological investigations, and hence the need for corroboration with genetic data. Accordingly, there is an urgency for research that focuses on the genetic and morphological characterization of Mt. Kenya's small mammals. This can effectively resolve their taxonomy while providing insights into community ecology along elevational gradients. This study analyzes the most abundant small mammals on Mt. Kenya—genus *Lophuromys*—with the objective to (i) investigate patterns of genetic (mitochondrial cytochrome-*b* gene) and morphological (external and craniodental) variation, (ii) confirm their taxonomic status using craniodental and phylogenetic data, and (iii) assess whether their morphology and genetics correlate with Mt. Kenya's elevational gradient, habitat structure, and climatic variation. These objectives also allowed us to infer morphological and genetic variation across an elevational gradient. These patterns can then be applied to relevant ecological hypotheses, including the thermoregulation hypothesis, where smaller body size is favored in warmer environments and vice versa (Olalla-Tarraga et al. 2006), and the resource availability hypothesis where conditions of low resource availability are hypothesized to constrain body size (Ernest et al. 2000).

Materials and methods

Study area and sampling methods

Small mammals were trapped on Mount Kenya National Park/Natural Forest between September and October 2015. Mount Kenya is the second highest mountain in Africa, with the highest peak rising to 5200 m. Sampling was conducted along the Chogoria (eastern/windward) and the Sirimon (western/leeward) slopes in stations set at approximately 200-m intervals along each slope. To capture animals, Sherman live traps, snap traps, and the sign trapping method (traps were laid at locations with active small mammal signs including droppings, trails, and burrows) were employed. For each specimen, tail length (TL), length of head and body (HB), length of hind foot (HF), and length of ears (EL) were measured in millimeters, along with mass in grams (WT). Due to the differential elevational distribution of habitats (vegetation stratification) on each slope, we only computed mean trap success by habitat. In addition to the Mt. Kenya specimens, we made craniodental measurements of the substantial collection of *Lophuromys* from the Aberdare Range (type locality of *Lophuromys zena*) and Mt. Kilimanjaro (type locality of *Lophuromys aquilus*) deposited at the Field Museum of Natural History (FMNH), Chicago, USA. The collection localities of samples used in the study are shown in Fig. 1 with details provided in Online Resource 1.

External morphological characterization

In addition to the external measurements, metrics of body surface area (BSA) and body mass index (BMI) were derived as both are integral traits underlining species distribution patterns. BSA is an important variable in mammalian energy utilization (Reynolds 1997) similar to BMI which was adopted as a measure of metabolic security and survival fitness (Padwal et al. 2016). BSA was calculated using the DuBois formula (Du Bois and Du Bois 1989): $BSA = (W^{0.425} \times HB^{0.725}) \times 0.007184$, where W = weight in kilograms and head-body (HB) length in centimeters. BMI was calculated as the ratio of body mass in grams and the squared value of head-body length (HB), $BMI = {}^{W}/_{HB^2}$ (Engelbregt et al. 2001). Principal components analysis (PCA) of the measurements' covariance matrix performed in PAST 3 (Hammer 2015) was used to obtain linearly uncorrelated values for subsequent analysis and to determine their variance contributions which was evaluated according to Hair et al. (1998). Non-parametric multivariate analysis of variance (NPMANOVA) (Anderson 2001) was used to determine the levels of statistical significance of differences. NPMANOVA allows greater adaptability in discerning group variations while offering ecological-friendly premises when dealing with multivariate datasets (Anderson 2001). The independent *t* test in SPSS (IBM Corp 2016) was used to determine the significance of univariate differences of group means. Statistical significances were determined from $p < 0.05$ values and were computed using the Mahalanobis similarity index, 10,000 permutation rounds, and the Bonferroni-type adjustment (Bland and Altman 1995) as performed in PAST. Only adult specimens (fully erupted M^3) were used, and all variables were log-transformed before undergoing statistical tests to enhance their multivariate dimensionality fit (McCune and Grace 2002). We also used histogram normality distribution curves and boxplots to identify and exclude outliers (unidentified non-adults and potential measurement errors). The raw values of external body measurements and derived metrics used are included in Online Resource 1.

Craniodental morphometric analysis

A total of 68 skulls (36 from Mt. Kenya, 13 from Mt. Kilimanjaro, and 19 from Aberdare) were used in the craniodental morphometric analyses. Linear measurements of 14 variables were obtained from the craniums using a digital calliper (Mitutoyo NTD12-6″ C) calibrated to the nearest hundredth (0.01) millimeter following Carleton and Van der Straeten (1997). Only adult specimens with fully erupted M^3 and visible molar wear were included. The measurements are illustrated in Fig. 2 and include condylo-incisive length (CI), greatest zygomatic breadth (ZB), breadth of braincase (BBC),

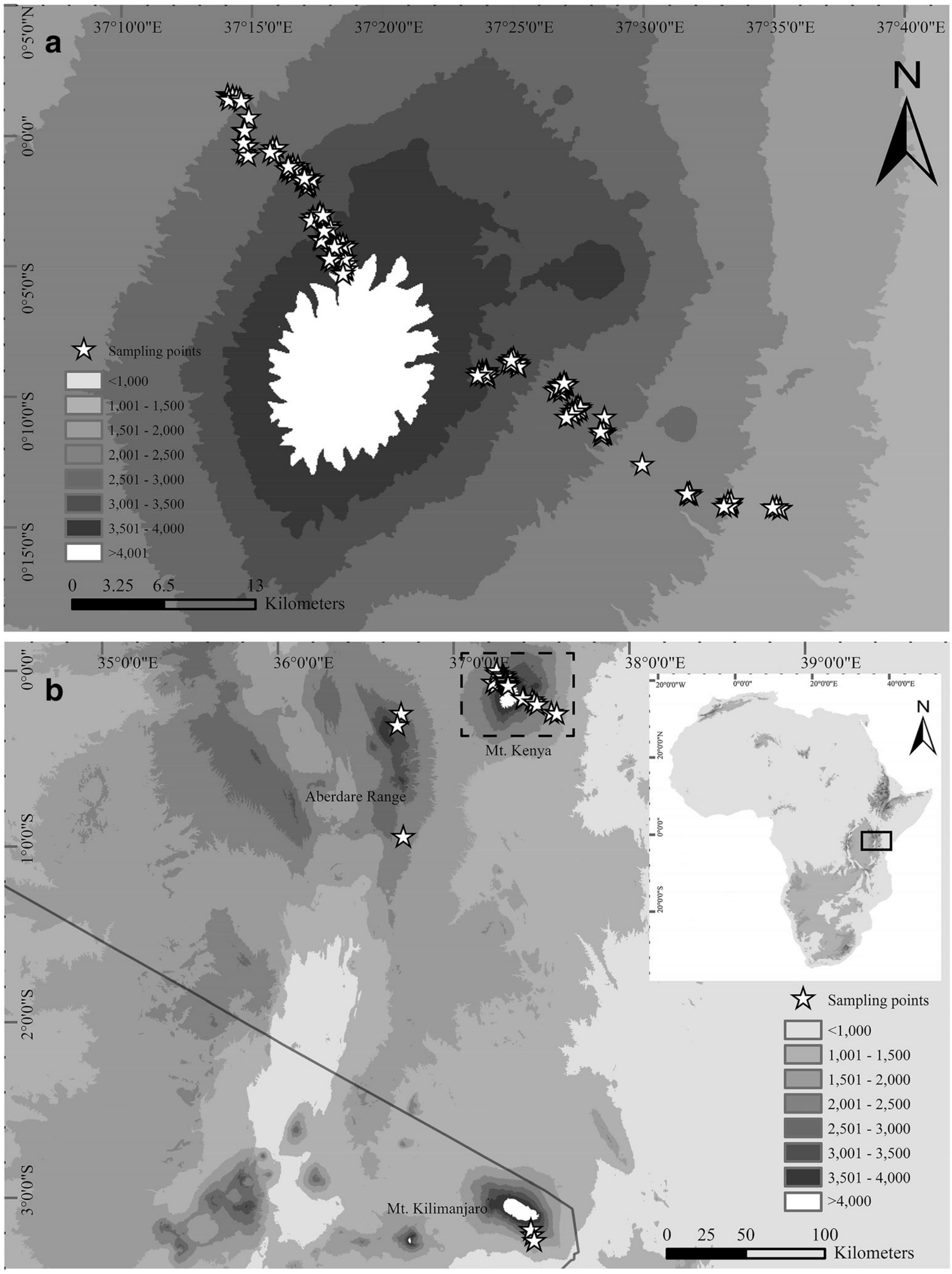

Fig. 1 Map showing the sampling localities of specimens used in the study (white-shaded stars); all sampling points of specimens used in the genetic and craniodental analyses (**a**) and sampling points on Mt. Kenya (**b**). The highlighted section in Fig. 1a inset (black outline rectangle) indicates the location of the study area within the African continent

breadth across the occipital condyles (BOC), least interorbital breadth (LOB), greatest breadth of rostrum (BR), post-palatal length (PPL), length of bony palate (LBP), length of incisive foramen (LIF), length of diastema (LD), breadth of zygomatic plate (BZP), length of auditory bulla (LAB), alveolar length of the maxillary tooth row (ALM), and coronal breadth of the first upper molar (BM1s). To ensure consistency, standardized orientation, and minimize errors, all measurements were made by KOO. The resulting values were log-transformed before being subjected to DFA and statistical significance tests. DFA, and not PCA, was used to explore group discriminations because it can estimate the combination of characters that best discriminate group differences (Strauss 2010; Viscosi and Cardini 2011). Otherwise, most of the statistical procedures of the craniodental dataset analyses were similar to those of the external morphological analyses above. The raw values of the linear measurements used for the craniodental analyses are included in Online Resource 1.

DNA extraction and PCR

Total DNA was extracted from muscle and/or liver tissue preserved at − 80 °C in 99.6% ethanol using the sodium dodecyl sulfate (SDS) method (Sambrook et al. 1989). One half

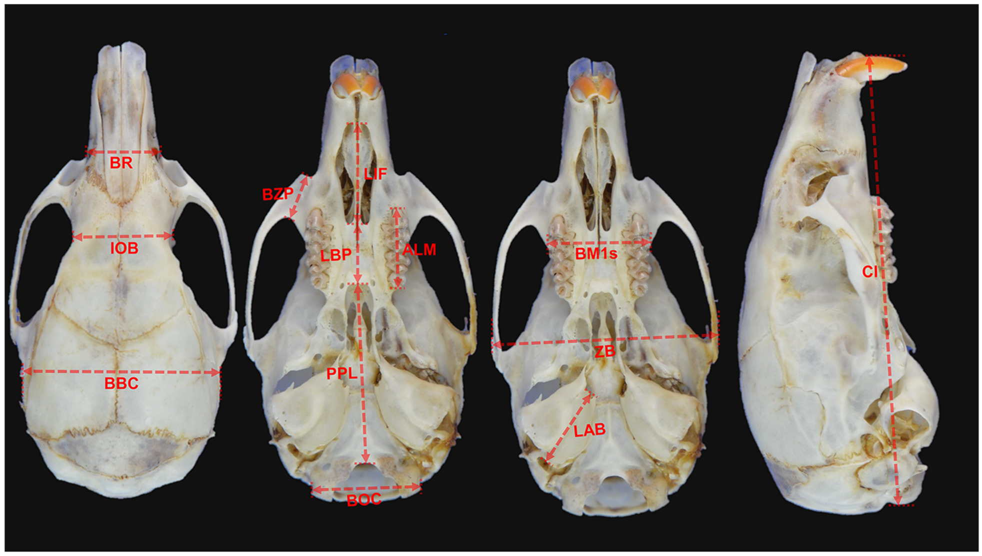

Fig. 2 Cranial pictures illustrating the 14 linear measurements used for the craniodental analyses

microliter of total DNA template was amplified by PCR using 0.5 μl of gene-specific primers 10 μl of 2× Power Taq PCR Master Mix, and 8.5 μl of ultrapure water to make a total volume of 20 μl reaction reagents per sample. A region of 1140 base pairs of the mitochondrial *Cytochrome-b* gene (*CYTB*) was successfully amplified using primers L14724 (5′-GGACTTATGACATGAAAAATCATCGTTG-3′) and H14139 (5′-GATTCCCCATTTCTGGTTTACAAGAC-3′) (Kocher et al. 1989). The PCR was run for 40 cycles consisting of denaturation at 95.0 °C for 30 s, annealing at 59.0 °C for 30 s, and extension at 72.0 °C for 60 s. The amplified product was then sequenced in forward and reverse directions using the ABI sequencer (Applied Biosystems). The sequences were assembled and manually edited using SeqMan in Lasergene7 (DNASTAR, Inc.; Madison, USA) and then aligned with the ClustalW algorithm in MEGA7 (Kumar et al. 2016).

Genetic diversity and phylogenetic analysis

We used the *CYTB* gene for inferring genetic diversity and phylogenetic relatedness due to its established relevance in mammalian phylogenetic reconstructions (Farias et al. 2001) and availability in the public repositories which enables comparisons with related lineages. The genetic dataset comprised 113 sequences in total, including 80 novel sequences generated from this study, 5 *L. stanleyi*, 3 *L. laticeps*, and 1 *L. zena* sequences downloaded from GenBank (Clark et al. 2016), and 24 *L. aquilus* sequences downloaded from the African Rodentia project (Terryn et al. 2007). We included the *L. stanleyi* and *L. laticeps* sequences because they were the only taxa available in GenBank that are close relatives of *L. zena* and were sampled from the Albertine Rift, which shares biogeographical affinities with our study area (Demos et al. 2014a, 2014b, 2015). We used DnaSP v5 (Librado and Rozas 2009) to assess population and individual-level genetic diversity in terms of the number of polymorphic sites, number of haplotypes, haplotype diversity, nucleotide diversity, and the average number of nucleotide differences for Mt. Kenya samples only. Haplotype structure, i.e., all shortest least complex haplotype relations, was inferred using the median joining network algorithm in PopART (Leigh and Bryant 2015). Maximum likelihood (ML) inference on the IQ-TREE web server (Trifinopoulos et al. 2016) using UFBoot (Hoang et al. 2017) and the TPM2u+F+I model (identified as the best-fit model according to BIC in ModelFinder; Kalyaanamoorthy et al. 2017) was used to estimate phylogenetic relationships. Pairwise estimation of between-group evolutionary divergence, i.e., the number of base substitutions per-site between groups, was analyzed using the Kimura 2-parameter model (Kimura 1980) in MEGA7. The dataset of unique Cytochrome-b sequences newly generated and analyzed in the current study are available in GenBank under accession numbers: [to be inserted upon acceptance of the manuscript].

Climatic influences

To determine whether ecosystem variables such as precipitation and temperature were significantly variable among habitats and elevations and were correlated with genetic and morphological variation, we performed correlation tests between the external body features and bioclimatic variables (extracted from Fick and Hijmans 2017) and elevation. The bioclimatic variables make up the common elements that impact the structuring of faunal community composition in montane ecosystems (McCain 2007). The Point Sampling Tool Plugin in QGIS (QGIS Development Team 2018) was used to cross-georeference specimen's latitude-longitude sampling coordinates in order to extract numerical values from publicly distributed bioclimatic variables (Fick and Hijmans 2017). From these 19 bioclimatic variables, only bio1, bio2, bio3, bio7, bio12, bio14, bio18, and bio19 were retained for subsequent analysis after removing highly correlated (> 0.8 Pearson's bivariate correlation) variables. The variables were log-transformed, and the PCA correlation matrix was used to explore the importance of their variance contributions. The PCA correlation matrix standardizes the different measurement scales to unit variances unlike the covariance matrix (Borgognone et al. 2001). The ratio of the variability of bioclimatic variables between elevations, habitat, and slopes was determined using one-way analysis of variance (ANOVA) in SPSS. Correlations between the bioclimatic variables and external body variables were explored using Pearson correlations in SPSS.

Results

Elevational distribution and abundance

From a total of 11,520 trap nights, 530 *Lophuromys* were collected, equalling 28% of all Mt. Kenya's snap trap and live trap captured small mammals. These included 277 (52.1%) males and 248 (46.6%) females, with 290 (54.5%) from Chogoria and 242 (45.5%) from Sirimon. Overall, *Lophuromys* captures were recorded in all sampling stations spanning the entire 1790–4200 m above mean sea level (m a.s.l.) elevation gradient. Peak trap success occurred at mid-elevations on both Chogoria and Sirimon slopes (Fig. 3a). Mean trap success varied from 1.9 to 7.7% between habitats (Fig. 3b). The mixed forest of bamboo and indigenous trees (2600–2800 m a.s.l.) had the highest trap success (7.7%) and the mixed indigenous forest (2000–2400 m a.s.l.) had the least trap success (1.9%).

External morphological description

Mount Kenya *Lophuromys* can be identified by the external body measurements in Table 1. A PCA conducted on these measurements to explore how they individually contributed to aggregate external morphological variation showed the first two components accounted for 93% (PC1 = 88% and PC2 = 5%) of the total variance. Weight and BSA had the most

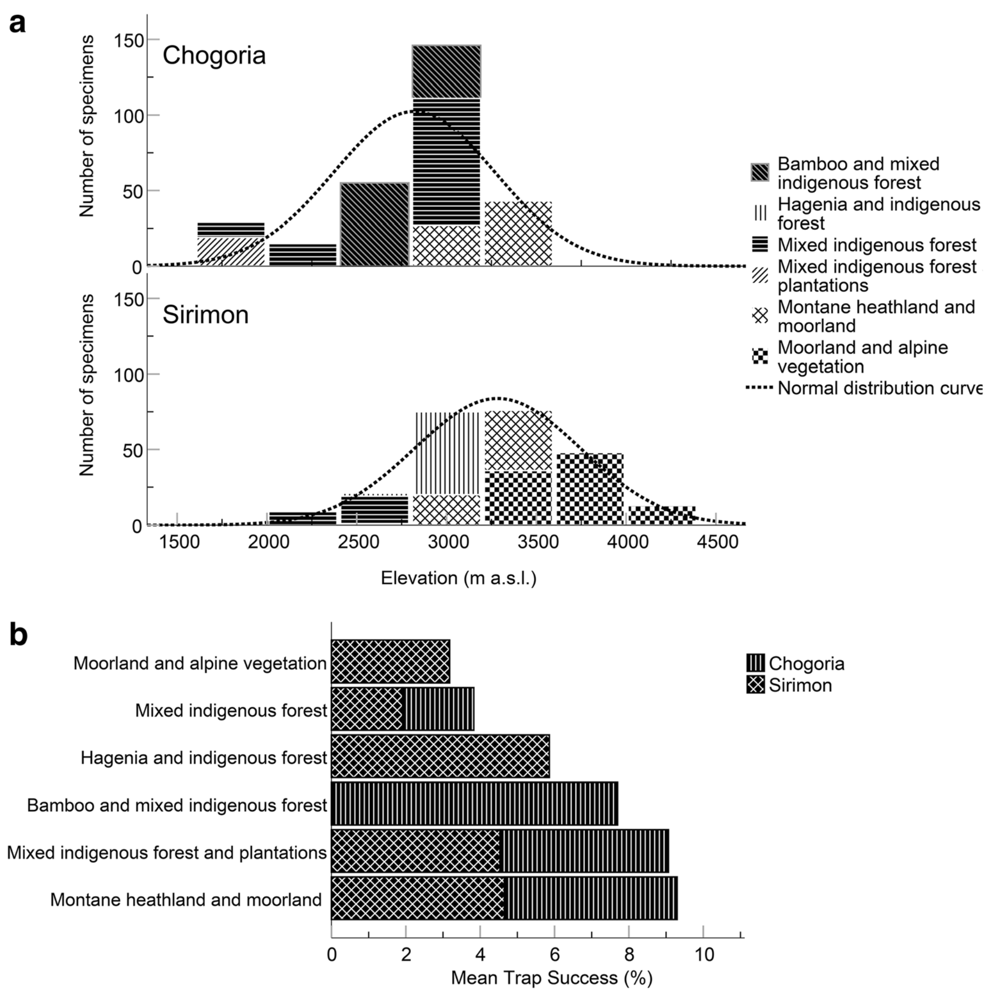

Fig. 3 Bar graphs showing the elevational occurrence of *Lophuromys* on Mt. Kenya; distribution by elevation and habitat (**a**) and mean trap success by habitat (**b**)

Table 1 Statistical metrics of the external body measurements of Mount Kenya *Lophuromys*. Only adult specimens (fully erupted M^3) included

Statistics	*M*	*SEM*	*SD*	S^2	*SK*	*R*	*Min*	*Max*	*N*
Weight (g)	47.72	0.36	8.31	68.98	0.05	46.90	24.80	71.70	527
Head-body (mm)	117.97	0.30	6.94	48.10	− 0.22	38.00	98.00	136.00	520
Tail (mm)	64.10	0.32	7.07	49.96	− 0.35	44.00	44.00	88.00	474
Tail/head-body × 100	54.68	0.26	5.68	32.27	− 0.73	34.34	35.16	69.49	474
Hind foot (mm)	20.51	0.04	0.95	0.91	0.28	5.50	18.00	23.50	526
Ear (mm)	18.71	0.04	0.90	0.81	0.45	5.00	17.00	22.00	516
BMI (g/cm^2)	0.3417	0.0016	0.0359	0.0013	0.2593	0.2146	0.2471	0.4618	521
BSA (cm^2)	117.88	0.57	12.99	168.79	− 0.12	71.06	81.97	153.02	521

M mean, *SEM* std. error of mean, *SD* std. deviation, S^2 variance, *SK* skewness, *R* range, *Min* minimum, *Max* maximum, *N* valid number of specimens

contribution of variance on PC1 (0.77 and 0.49, respectively), an indication that the first principal component represented the overall body size. The second component was mostly associated with BMI (− 0.83) and head-body (0.42), suggesting that surface area to volume ratio is an important trait for the Mount Kenya *Lophuromys*. Ear and hind foot length had overall low contributions to variances on PC1 (0.08 and 0.06, respectively) and PC2 (0.09 and 0.11, respectively).

Mt. Kenya *Lophuromys* are significantly sexually dimorphic based on external body measurements. A non-parametric multivariate analysis of variance showed that ♂s and ♀s were distinguishable by external body measurements in a combined dataset ($p < 0.0001$, $F_{5,445}$ 9.411), as well as individually in Chogoria and Sirimon ($p < 0.0001$, $F_{5,237} = 5.806$, and $p < 0.0001$, $F_{5,202} = 11.73$, respectively). Independent *t* test comparisons showed significant sexual dimorphism on PC1 and PC2 in the combined dataset. Further comparative analyses indicated that Chogoria specimens were significantly sexually dimorphic on both PC1 and PC2 while those from Sirimon were sexually dimorphic on PC1 scores only (Fig. 4). Variable-specific comparisons showed that ♂s were overall heavier and larger than ♀s regardless of the sampled slope (Online Resource 3). By vegetation clusters, specimens were significantly distinct (between groups ANOVA $p < 0.05$ for all measurements, with the exception of ear length and BMI). Specimens from the indigenous forest mixed with plantations were the largest and heaviest, followed by mixed indigenous forest, mixed bamboo and indigenous forest, montane heathland and moorland, moorland and alpine vegetation, and mixed Hagenia and indigenous forest, in respective order of decreasing body size and weight (Online Resource 4).

Craniodental morphometric characterization

The craniodental analyses revealed that all three populations were not sexually dimorphic; Mt. Kenya ($p = 0.7138$, $F_{33.33,32.49} = 0.8198$), Mt. Kilimanjaro ($p = 0.7142$, $F_{9.641,8.986} = 0.7283$), and Aberdare ($p = 0.1999$, $F_{14,13} = 1$). Both males and females were therefore combined for subsequent analyses. There was an overall statistically significant craniodental difference between the three populations ($p < 0.001$, $F_{62.14,56.63} = 2.918$). Pairwise multivariate tests showed that only Mt. Kilimanjaro's population was significantly distinct from the Mt. Kenya ($p < 0.001$) and Aberdare ($p < 0.001$) populations and that Mt. Kenya and Aberdare populations were not significantly differentiated ($p =$ 0.0564). Univariate comparisons indicated that *Lophuromys* of Mt. Kilimanjaro had larger skulls than those on Mt. Kenya and Aberdare while those of Mt. Kenya and Aberdare had similar mean craniodental values (Table 2). DFA analysis

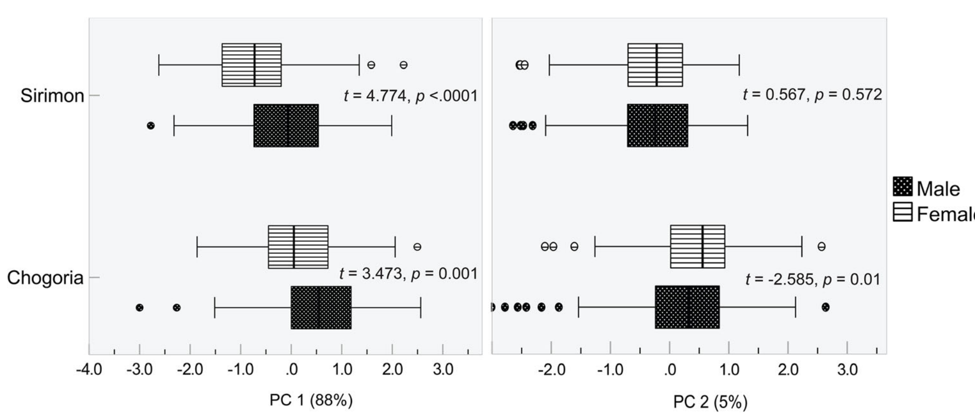

Fig. 4 Boxplots exploring sexual dimorphism of external body measurements of Mt. Kenya's *Lophuromys* by slope. The independent *t* test results and associated *p* values are shown for each comparison. The lines inside boxes indicate median, whiskers indicate minimum and maximum values excluding outliers (circles), and boxes' lower and upper bounds indicate the 25th and 75th percentiles, respectively

Table 2 Univariate mean comparisons of linear craniodental variables of *Lophuromys* from Mt. Kenya, Mt. Kilimanjaro, and Aberdare

Variable	Mt. Kilimanjaro			Mt. Kenya			Aberdare		
	$\bar{x}$	*SD*	*N*	$\bar{x}$	*SD*	*N*	$\bar{x}$	*SD*	*N*
CI	29.26	0.45	13	28.49	1.05	36	28.08	0.73	19
ZB	14.87	0.34	13	14.90	0.50	36	14.69	0.60	19
BBC	12.98	0.24	13	12.72	0.32	36	12.72	0.25	19
PPL	12.48	0.25	13	11.99	0.49	36	11.58	0.51	19
LD	8.73	0.18	13	8.24	0.39	36	8.11	0.34	19
BOC	7.19	0.17	13	7.26	0.23	35	7.10	0.30	19
BM1s	6.72	0.09	13	6.90	0.23	36	6.69	0.22	19
LIF	6.74	0.28	13	6.50	0.30	36	6.45	0.32	19
IOB	6.05	0.12	13	6.08	0.22	36	5.99	0.24	19
LAB	5.48	0.12	13	5.57	0.19	36	5.44	0.25	19
ALM	4.72	0.14	13	4.82	0.20	36	4.90	0.19	19
BR	5.06	0.18	13	4.98	0.23	36	4.73	0.24	19
LBP	4.15	0.21	13	4.23	0.31	36	4.18	0.19	19
BZP	2.80	0.10	13	3.00	0.17	36	2.99	0.19	19

correctly predicted all the Mt. Kilimanjaro specimens, whereas specimens from Mt. Kenya and Aberdare Range could not be predicted. A scatter plot of the first DFA axis (84.9%) compared to the second axis (15.1%) showed that Mt. Kilimanjaro specimens cluster distinctly in morphospace to the exclusion of specimens from Mt. Kenya and Aberdare (Fig. 5). Separately, the Mt. Kenya specimens did not have significant among group divergence between vegetation clusters ($p = 0.257$, $F_{35.27,29.82} = 1.098$) and elevation intervals ($p = 0.3746$, $F_{33.25,30.08} = 1.054$).

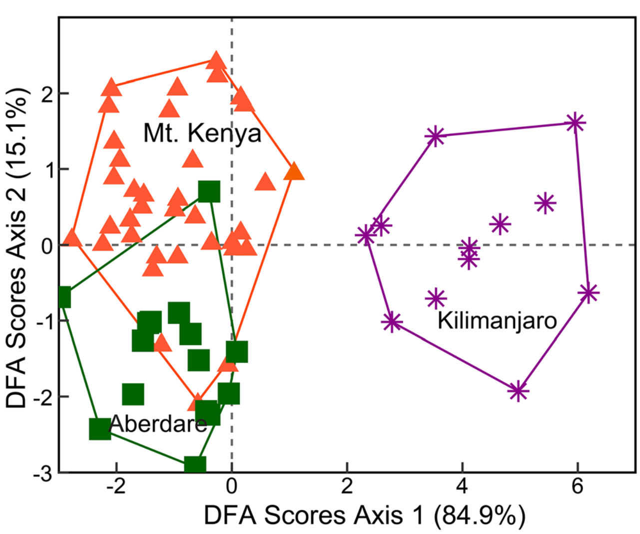

Fig. 5 Bivariate scatter plot of the first and second DFA scores of linear craniodental variables of *Lophuromys specimens* from Mt. Kenya, Mt. Kilimanjaro, and Aberdare

Genetic diversity and phylogeny

Genetic diversity tests showed low *CYTB* variation within Mt. Kenya *Lophuromys* specimens (3.34% polymorphic sites, 0.0043 nucleotide diversity) and high haplotype diversity (0.93). Twenty-two haplotypes, 38 polymorphic sites, 4.8 average number of nucleotide differences, and 0.004 average evolutionary divergences were detected. Pairwise between-group genetic dissimilarity distances were low between Mt. Kenya and Aberdare (0.0065), but high between Kilimanjaro and Mt. Kenya (0.1020), and Kilimanjaro and Aberdare (0.1020), affirming that central Kenya *Lophuromys* is genetically differentiated from the Mt. Kilimanjaro population. In comparison, *L. stanleyi* was genetically distant from both Mt. Kenya and Aberdare populations (0.0308 and 0.0317, respectively), as was *L. laticeps* (Mt. Kenya, 0.0413; Aberdare, 0.0417). The pairwise Kimura 2-parameter model distance between *L. stanleyi* and *L. laticeps* (0.0381) was about the same distance between them (*stanleyi* and *laticeps*) and both Mt. Kenya and Aberdare samples.

The *CYTB* ML phylogeny supported the Mt. Kenya and Aberdare *Lophuromys* populations as monophyletic and distinctly separated from the monophyletic Mt. Kilimanjaro *Lophuromys* population (Fig. 6a). The terminal branching of the Mt. Kenya (Chogoria and Sirimon) and Aberdare clade suggested no apparent pattern either by slope or elevation. Median-joining haplotype networks also established that Mt. Kilimanjaro *Lophuromys* cluster as a single population, separated by several mutations from the Mt. Kenya and Aberdare haplotypes (Fig. 6b) which also cluster as a common population. The Mt. Kenya haplotypes comprised samples from different habitats (Fig. 7a) and elevations (Fig. 7b), indicating that neither elevational variation nor vegetation stratification influenced the genetic structure of *Lophuromys*. Overall, both the haplotype networks and ML phylogenetic results delimitate the central Kenya (Mt. Kenya and Aberdare) populations from the Mt. Kilimanjaro *Lophuromys* population.

Climatic influences

The bioclimatic variables were significantly variable between vegetation clusters, elevation intervals, and slopes (all ANOVA $p < 0.0001$), suggesting that they interacted with the craniodental and external morphology and genetic variation in *Lophuromys*. The PCA of the bioclimatic variables showed PC1 highly associated with annual mean temperature (bio1), precipitation of the driest month (bio14), and precipitation of coldest quarter (bio19). Annual precipitation (bio12), precipitation of the driest month (bio14), and precipitation of the warmest quarter (bio18) had very high contributions on PC2 while precipitation of the coldest quarter (bio19) contributed most on PC3. With the exceptions of mean diurnal temperature range, isothermality, and temperature annual range,

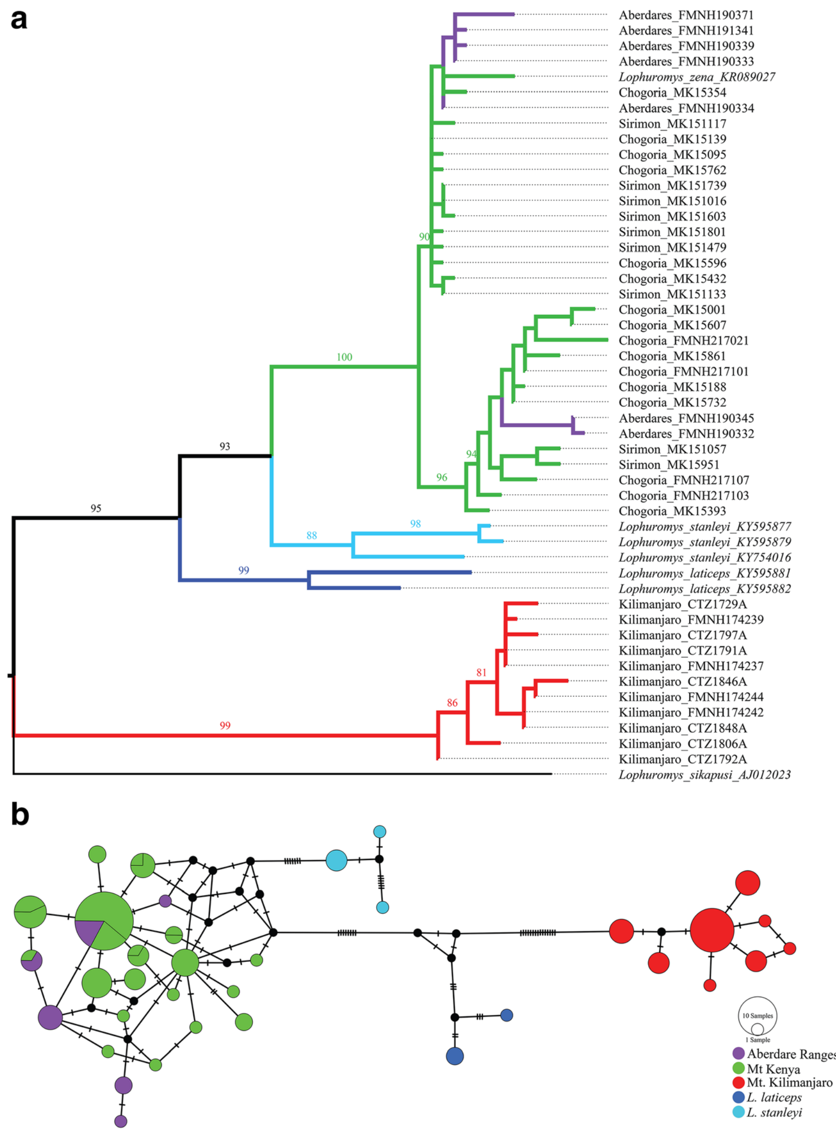

Fig. 6 Phylogenetic and population genetic analysis of Mt. Kenya, Aberdare, and Mt. Kilimanjaro *Lophuromys aquilus* species and *L. stanleyi* and *L. laticeps* using *CYTB*. The maximum likelihood phylogeny was inferred using only unique haplotypes in IQ-TREE (**a**). Bootstrap support for major branches only are represented by the values above branches. Haplotype networks inferred using the Median-Joining algorithm in PopART (**b**). Branch hatch marks represent the number of mutations between haplotypes and small black circles represent missing haplotypes

all the bioclimatic variables had contributions less than − 0.3 or > 0.3 on the first three principal components (which altogether accounted for 89% of the total variance). BSA (adopted as a surrogate of weight and head-body) was negatively correlated with elevation, temperature annual range, and precipitation of coldest quarter, but positively with all remaining bioclimatic variables. BMI was negatively correlated with all bioclimatic variables except annual mean temperature, mean diurnal temperature range, isothermality, and elevation.

Discussion

Taxonomic insight into Mt. Kenya *Lophuromys*

Our findings have established that Mt. Kenya *Lophuromys* exhibit no consistent patterns in morphological variation (external body and craniodental measurements) or genetics associated with altitude, vegetation, or climatic variables. Further, these results demonstrate that the Mt. Kenya and Aberdare populations of *Lophuromys* are indistinguishable by craniodental and genetic variation, suggesting that they belong to the same species and that this species is evolutionarily distinct from the Mt. Kilimanjaro *Lophuromys* population. Integrative taxonomy indicates that the Mt. Kenya *Lophuromys* population represents *L. zena* (Dollman 1909). This occurrence of a single species widely distributed in a mountain ecosystem is not unparalleled as other shrew and rodent species in the region have portrayed similar distribution patterns. In their characterization of Mt. Kilimanjaro *Lophuromys* using craniometrical analyses, Verheyen et al. (2007) did not uncover consistent characters that discriminated individuals among different elevations, and accordingly, they should be inferred as a single taxonomic unit (*L. aquilus*). Other taxa that have been recorded with similar single-species occurrences on mountains despite considerable

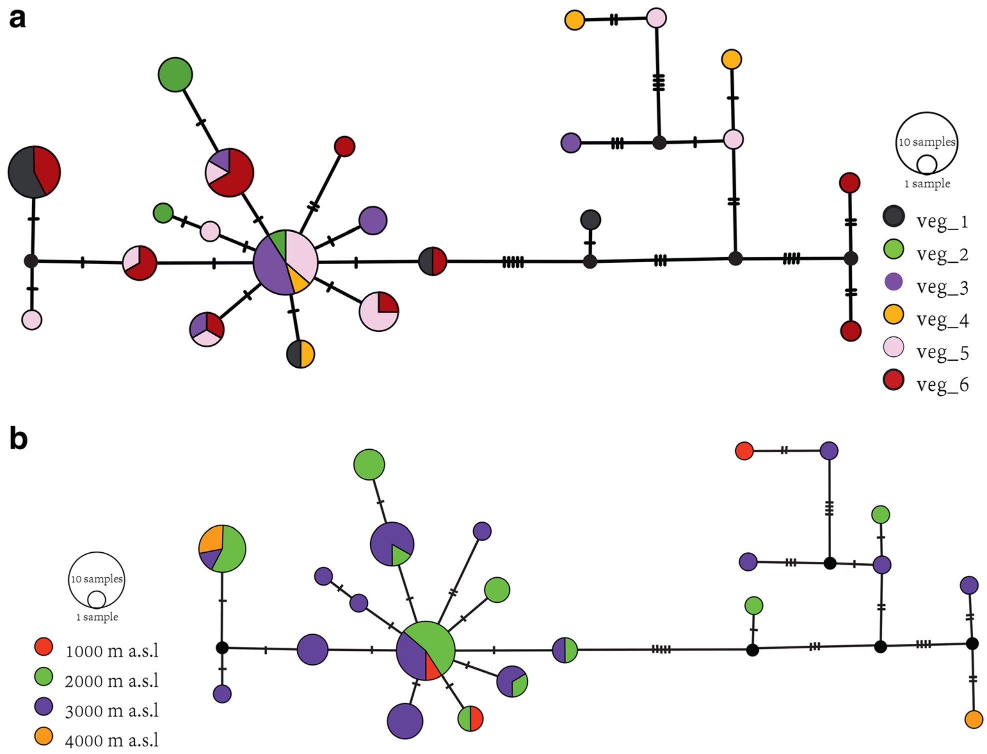

Fig. 7 *CYTB* haplotype networks of Mt. Kenya *Lophuromys* specimens inferred using the median-joining algorithm in PopART, shaded by habitats: (**a**) veg_1: bamboo and mixed indigenous forest, veg_2: Hagenia and indigenous forest, veg_3: mixed indigenous forest, veg_4: mixed indigenous forest and plantations, veg_5: montane heathland and moorland, veg_6: moorland and alpine vegetation, and elevation (**b**). Branch hatch marks represent the number of mutations between haplotypes and small black circles represent missing haplotypes

elevational stratifications of ecological variables include *Surdisorex polulus* and *Praomys jacksoni* of Mt. Kenya (Musila et al. 2019), *Surdisorex norae* of Aberdare Range (Peterhans et al. 2009), *Hylomyscus endorobae* of Mt. Kenya and Aberdare (Demos et al. 2014a, 2014b), and *Myosorex zinki* and *Praomys delectorum* of Mt. Kilimanjaro (Stanley et al. 2014). Nonetheless, there have been reports of single mountain elevational parapatry of *Lophuromys* species, including Lavrenchenko et al. (1998) and Lavrenchenko et al. (2000) who established that *L. brevicaudas*, *L. chrysopus*, and *L. melanonyx* all occurred in the Bale Mountains, Ethiopia. Kostin et al. (2019) also recorded *L. brevicaudas*, *L. chrysopus*, and *L. melanonyx* with a similar occurrence pattern in the Arsi Mountains, Ethiopia. In both mountains, the species replaced each other across elevation and were associated with discrete habitat types, a scheme which Lavrenchenko et al. (1998) attributed to an adaptive speciation model.

Our results, which show similar levels of intra- and inter-group genetic variation between the Mt. Kenya-Aberdare *Lophuromys* population and *L. stanleyi* as reported in Verheyen et al. (2002) and Verheyen et al. (2007), suggest that *L. zena* is not endemic to the montane habitats of Aberdare Range and Mt. Kenya regions, but instead has a broader distribution. This is supported by the absence of any imposing biogeographical barrier (in terms of terrain, elevation, or climate) between Mt. Kenya, Aberdare, Loita, Mau, and Kakamega localities, from where results of this study and other data (unpublished) indicate that they are occupied by *Lophuromys* populations indistinguishable morphologically and genetically. Verheyen et al. (2007) did not present phylogeographic or historical biogeographic evidence that *Lophuromys* specimens from Solai, Mt. Elgon, and Cherangani represent *L. margarettae*, whereas those from Kaptagat represent *L. zena*. This hypothesis is a rather unlikely zoogeographic pattern considering that Solai, for instance, is more than twice as close to Aberdare, the type locality of *L. zena*, than Kaptagat. In effect, we are unable to find explanations for Verheyen et al. (2002) and Verheyen et al. (2007) definitive conclusions that *Lophuromys* from lower elevations of the Aberdare Range and Mt. Kenya are allocated to "*aquilus*" (i.e., *L. margarettae*) whereas individuals from higher elevations belong to *L. zena*. Clearly, additional analyses using samples from all the Kenyan forests are necessary for a resolute conclusion on the distributional and taxonomic limits of *L. zena* and *L. margarettae*.

Ecology of the Mt. Kenya *Lophuromys*

The high abundance and wide distribution of *Lophuromys* on Mt. Kenya is consistent with previous surveys of the Eastern Africa mountains (Rwenzori Mountains (Peterhans et al. 1998), Mt. Elgon (Clausnitzer and Kityo (2001), Mt. Kilimanjaro (Mulungu et al. 2008)). However, the elevational gradient spanned by the Mt. Kenya population (1790–4100 m a.s.l.) includes a wide range in temperature, precipitation, and solar exposure, among other ecological elements which often involve cryptic adaptive morphological speciation (Wells 2007). Our findings indicate that *Lophuromys* is a highly successful rodent within the Mt. Kenya ecosystem (by occurrence and abundance) with no evidence of morphological or genetic partitioning to suggest the presence of congeneric populations. The significant but random pattern of variation in external body characters and genetics by elevation or habitat and their elevational occurrence can consequently be ascribed to several ecological correlations. For instance, on the basis of *Lophuromys* mostly insectivore diet, which Dieterlen (1976) attributed to their demographic success even in harsh alpine zones, the wide elevational occurrence on Mt. Kenya may be associated with the distribution and abundance of invertebrates (Sømme and Zachariassen 1981). Likewise, the overall *CYTB* homogeneity on Mt. Kenya supports this *Lophuromys* population as panmictic. Support for this includes low genetic diversity and lack of population genetic structure (Crandall et al. 2000, Hanski 2013). In addition, the observed population structure (or lack thereof) may result from recent population expansion leading to a scenario where all individuals within a population coalesce on a recent common ancestor (Avise 2000).

Small mammals such as rodents generally have a higher BSA to volume ratio which increases their potential heat loss in cold environments and heat gain in hot environments, promoting more effective thermoregulatory adaptations. Accordingly, the negative correlation between BSA and elevation among Mt. Kenya *Lophuromys* contradicts the heat dissipation (thermoregulation) hypothesis—that higher rates of heat loss favor small-bodied mammals in warm conditions (lower elevation) while individuals at higher elevations require larger BSA to minimize unstable heat dissipation. However, the productivity hypothesis—that conditions of resource scarcity (typical of higher elevations) constrain species overall growth (body size)—fits the observed correlation between BSA and elevation. The positive correlation between BMI and bioclimatic temperature variables and negative correlation with precipitation variables also suggests that Mt. Kenya *Lophuromys* are under stronger selection for heat regulation than water limitations.

In general, the bioclimatic variables associated with precipitation appear to be better predictors of habitat suitability for Mt. Kenya *Lophuromys* than those associated with temperature, which is in agreement with Dieterlen (1976) who found that the distribution of *Lophuromys,* across most of its range, is primarily influenced by precipitation. The nonsignificant multivariate differences in external body measurements between habitats indicate that *Lophuromys* external morphology is not substantially influenced by the distinct vegetation stratification on Mt. Kenya. The univariate differences (Online Resource 3) are thus attributable to other ecological factors, such as variable resource availability along elevation gradients of tropical mountains (McCain 2005). On the other hand, the significant differences in external morphology between slopes, elevation, and sexes suggest that populations have differing responses to environmental variation across the mountain. Although the absence of craniodental sexual dimorphism contrasts with external morphological dimorphism, it is in agreement with previous studies (Bekele and Corti 1994; Verheyen et al. 1996). Similarly, the lack of significant differences in craniodental variables between sexes, vegetation clusters, and elevation intervals suggest *Lophuromys* craniums, unlike their external body morphology, are less plastic to environmental selection processes facilitating species diagnoses. Notably, the high degree of pelage coloration polymorphism observed among Mt. Kenya *Lophuromys* appears to be a polymorphic trait that may not be under sexual selection, making it an unsuitable trait to rely on for species delimitation.

Conclusion

In conclusion, this study investigated the morphology (craniodental and external characters) and genetics (*CYTB*) of the most abundant and widely distributed small mammal on Mount Kenya—genus *Lophuromys*—to deduce associations between taxonomy and elevational ecology. We found no consistent patterns in the variation of these variables by elevation, gradient, or habitat, suggesting that the population of brush-furred rats on Mount Kenya is a single species along the entire elevation gradient (1800–4100 m a.s.l). In addition, the study found significant variation in external morphology but not in craniodental morphology between elevations, sexes, and slopes (Chogoria and Sirimon), suggesting that patterns of variability in these skull morphology should not be interpreted as responding to an elevation gradient. The Mt. Kenya *Lophuromys* distribution overlaps in craniodental and genetic characters with the Aberdare *L. zena*, and both populations are both distant from the Mt. Kilimanjaro *L. aquilus*. Consequently, our findings support all Mt. Kenya *Lophuromys* as a single population, minimally differentiated from the Aberdare Range population, and assignable to *Lophuromys zena*. This taxonomic conclusion conflicts with that of Verheyen et al. (2002) which hypothesized the occurrence of *L. margarettae* at lower elevations and *L. zena* at higher elevations of Mt. Kenya. However, we find no morphological or genetic grounds for such a claim. The possibility that *L. margarettae* might be distributed in the Mt. Kenya region, although highly unlikely, can only be ascertained by further sampling in habitats lower than the 1700–4100 m gradient covered in this study. The *Lophuromys aquilus* species group appears to be in need of further taxonomic and distributional clarification to resolve existing discrepancies. While the precise ecological responses of species to changing climates and human interactions remain largely unknown for most rodent species (Askay et al. 2014), understanding locally abundant taxa such as *Lophuromys* can provide a much-needed baseline to inform more adept biodiversity conservation responses. The factors underlining *Lophuromys* high abundance and wide elevational gradient distribution on Mount Kenya could lead to a better understanding of the responses of other species to climate change (Beniston 2003; Boutin and Lane 2014) and habitat destruction (Eckert et al. 2017). As such, future studies could refine such approaches by investigating epigenetic adaptations of locally abundant small mammal taxa along elevational gradients.

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Compliance with ethical standards

Ethical approval All procedures involving animals adhered to the wildlife research laws and regulations of Kenya and Tanzania and the guidelines for the use of wild animals in research and education as in Sikes and The Animal Care and Use Committee of the American Society of Mammologists (2016).

Conflict of interest The authors declare that they have no conflict of interest.

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